Class 10 Mathematics (NCERT) – Chapter 10: Circles

Worked solutions: Examples 1–3, Exercise 10.1 and Exercise 10.2

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Properties and formulas used in this chapter

(P1) **Tangent–radius property.** The tangent to a circle at a point is perpendicular to the radius through the point of contact.

(P2) **Equal-tangents property.** The lengths of the two tangents drawn from an external point to a circle are equal.

(P3) **Centre on the angle bisector.** If two tangents are drawn from an external point, the line joining that point to the centre bisects the angle between the tangents. (Proof: the two right triangles formed with the centre are congruent by RHS.)

(P4) **Perpendicular from the centre to a chord** bisects the chord.

(P5) **Pythagoras theorem.** In a right triangle, $(\text{hypotenuse})^2 = (\text{base})^2 + (\text{perpendicular})^2$.

(P6) **Angle-sum property.** The angles of a triangle add up to 180°; the angles of a quadrilateral add up to 360°.

(P7) **Heron's formula** (used in Ex. 10.2 Q12). For sides a, b, c and $s = \dfrac{a+b+c}{2}$, area $= \sqrt{s(s-a)(s-b)(s-c)}$.

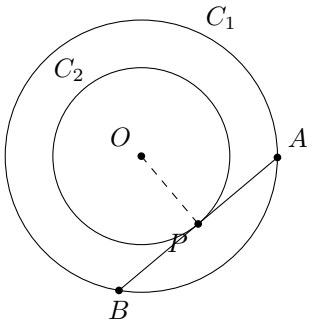
(P8) **Area of a triangle with inradius r.** Area $= \dfrac{1}{2} \times r \times (\text{perimeter})$.

Layout of a proof. Every proof is written as Given, To prove, Construction (if needed), then a Step / Statement / Reason table. The reason column names the property used in words.

Part A: Solved Examples 1 to 3

Example 1

Question. Prove that in two concentric circles, the chord of the larger circle, which touches the smaller circle, is bisected at the point of contact.



Concentric circles C_1 (larger) and C_2 (smaller); chord AB of C_1 touches C_2 at P. Dashed line: construction.

Solution.

Given: Two concentric circles C_1 and C_2 with centre O. AB is a chord of the larger circle C_1 and touches the smaller circle C_2 at P.

To prove: $AP = BP$.

Construction: Join OP.

Proof:

Step	Statement	Reason
1	AB is a tangent to C_2 at P and OP is the radius of C_2 through P, so $OP \perp AB$.	The tangent to a circle is perpendicular to the radius through the point of contact.
2	AB is a chord of C_1 and $OP \perp AB$, where O is the centre of C_1.	The perpendicular from the centre of a circle to a chord bisects the chord.
3	$AP = BP$.	From step 2.

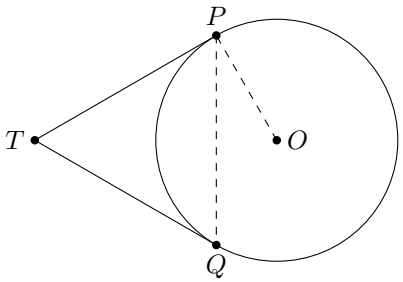
Common mistake. *The perpendicular is drawn from the centre O of the* larger *circle, and the radius OP used is the radius of the* smaller *circle (it ends at the point of contact). Do not mix up which circle each fact refers to.*

Answer: $AP = BP$, so the chord AB is bisected at the point of contact P. (Proved)

Alternative proof (without the chord property). In $\triangle OPA$ and $\triangle OPB$: $\angle OPA = \angle OPB = 90°$ (step 1), $OA = OB$ (radii of C_1), and OP is common. So $\triangle OPA \cong \triangle OPB$ (RHS), and $AP = BP$ (CPCT).

Example 2

Question. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$.



TP and TQ are tangents; P and Q are the points of contact. Dashed lines: construction.

Solution.

Given: A circle with centre O; an external point T; tangents TP and TQ with points of contact P and Q.

To prove: $\angle PTQ = 2\angle OPQ$.

Construction: Join PQ and OP.

Proof: Let $\angle PTQ = \theta$.

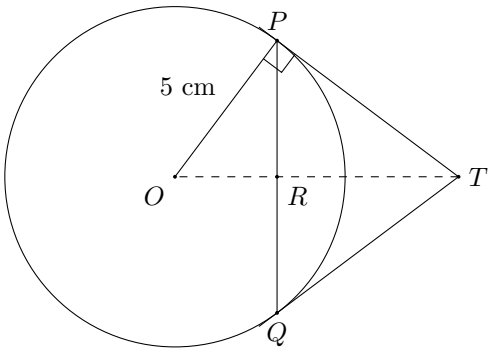
Step	Statement	Reason
1	$TP = TQ$.	The lengths of the two tangents from an external point are equal.
2	$\angle TPQ = \angle TQP$.	Angles opposite equal sides of $\triangle TPQ$ are equal.
3	$\angle TPQ + \angle TQP + \theta = 180°$.	Angle-sum property of $\triangle TPQ$.
4	$\angle TPQ = \frac{1}{2}(180° - \theta) = 90° - \frac{1}{2}\theta$.	From steps 2 and 3.
5	$\angle OPT = 90°$.	The tangent is perpendicular to the radius through the point of contact.
6	$\angle OPQ = \angle OPT - \angle TPQ$.	Ray PQ lies inside $\angle OPT$.
7	$\angle OPQ = 90° - \left(90° - \frac{1}{2}\theta\right) = \frac{1}{2}\theta$.	From steps 4, 5 and 6.
8	$\theta = 2\angle OPQ$.	From step 7.

Common mistake. *In step 6 the angle $\angle OPQ$ is the* difference $\angle OPT - \angle TPQ$*, not the sum. Also, $\angle OPT = 90°$ only because OP is a radius and TP a tangent at the same point P.*

Answer: $\angle PTQ = 2\angle OPQ$. (Proved)

Example 3

Question. PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at a point T. Find the length TP.



Chord $PQ = 8$ cm, radius $OP = 5$ cm; tangents at P and Q meet at T. Dashed line: construction (OT meets PQ at R).

Solution. Join OT and let it meet PQ at R.

Step 1: R is the mid-point of PQ. $TP = TQ$ (tangents from an external point are equal), so $\triangle TPQ$ is isosceles, and TO bisects $\angle PTQ$ (the centre lies on the bisector of the angle between the tangents). The bisector of the vertex angle of an isosceles triangle is perpendicular to the base and bisects it. Hence $OT \perp PQ$ and

$$PR = RQ = \tfrac{1}{2} \times 8 = 4 \text{ cm}.$$

Step 2: find OR. In the right triangle ORP ($\angle ORP = 90°$), by the Pythagoras theorem $OP^2 =$

$OR^2 + PR^2$:

$$OR^2 = OP^2 - PR^2 = 5^2 - 4^2 = 25 - 16 = 9$$
$$OR = \sqrt{9} = 3 \text{ cm}.$$

Common mistake. *The hypotenuse of $\triangle ORP$ is OP (opposite the right angle at R), so OP^2 is the sum, and $OR^2 = OP^2 - PR^2$ is a difference. Writing $5^2 + 4^2$ here is wrong.*

Step 3: two equal angles (this answers the book's "Why?"). The tangent is perpendicular to the radius, so $\angle TPO = 90°$, i.e.

$$\angle TPR + \angle RPO = 90°.$$

In $\triangle TRP$, $\angle TRP = 90°$, so by the angle-sum property

$$\angle TPR + \angle PTR = 90°.$$

Comparing the two, $\angle RPO = \angle PTR$.

Step 4: similar triangles. In $\triangle TRP$ and $\triangle PRO$: $\angle TRP = \angle PRO = 90°$ and $\angle PTR = \angle RPO$ (Step 3). So $\triangle TRP \sim \triangle PRO$ (AA similarity), and corresponding sides are proportional:

$$\frac{TP}{PO} = \frac{RP}{RO} \;\Rightarrow\; \frac{TP}{5} = \frac{4}{3} \;\Rightarrow\; TP = \frac{20}{3} \text{ cm}.$$

Verification. In the right triangle TRP (right angle at R), by the Pythagoras theorem $TR^2 = TP^2 - PR^2$:

$$TR^2 = \left(\frac{20}{3}\right)^2 - 4^2 = \frac{400}{9} - \frac{144}{9} = \frac{256}{9}, \qquad TR = \frac{16}{3} \text{ cm}.$$

So $OT = TR + RO = \dfrac{16}{3} + 3 = \dfrac{25}{3}$ cm. In $\triangle OPT$, by the Pythagoras theorem $OT^2 = TP^2 + OP^2$ should hold:

$$TP^2 + OP^2 = \left(\frac{20}{3}\right)^2 + 5^2 = \frac{400}{9} + \frac{225}{9} = \frac{625}{9}, \qquad OT^2 = \left(\frac{25}{3}\right)^2 = \frac{625}{9}.$$

Both sides are equal, so the answer is correct.

Answer: $TP = \dfrac{20}{3}$ cm $\left(= 6\frac{2}{3} \text{ cm}\right)$.

Alternative solution (Pythagoras theorem). Let $TP = x$ and $TR = y$, so $OT = TR + RO = y + 3$.

In the right triangle PRT (right angle at R), $TP^2 = TR^2 + PR^2$ with $PR = 4$:

$$x^2 = y^2 + 4^2 = y^2 + 16.$$

In the right triangle OPT (right angle at P), $OT^2 = TP^2 + OP^2$ with $OP = 5$:

$$(y + 3)^2 = x^2 + 5^2 = x^2 + 25.$$

Expand the left side by the identity $(a + b)^2 = a^2 + 2ab + b^2$, giving $(y + 3)^2 = y^2 + 2 \cdot y \cdot 3 + 3^2 = y^2 + 6y + 9$, and put $x^2 = y^2 + 16$ on the right:

$$y^2 + 6y + 9 = (y^2 + 16) + 25$$
$$y^2 + 6y + 9 = y^2 + 41$$
$$6y = 41 - 9 = 32$$
$$y = \frac{32}{6} = \frac{16}{3}.$$

Then, putting $y = \dfrac{16}{3}$ and writing $16 = \dfrac{144}{9}$:

$$x^2 = \left(\frac{16}{3}\right)^2 + 16 = \frac{256}{9} + \frac{144}{9} = \frac{400}{9}, \qquad x = \sqrt{\frac{400}{9}} = \frac{20}{3} \text{ cm}.$$

Part B: Exercise 10.1

Exercise 10.1, Question 1

Question. How many tangents can a circle have?

Solution. At every point of a circle there is one and only one tangent: it is the line drawn through that point perpendicular to the radius to that point (the tangent is perpendicular to the radius through the point of contact). A circle has infinitely many points, so it has infinitely many tangents.

Answer: A circle can have infinitely many tangents.

Exercise 10.1, Question 2 (i)

Question. Fill in the blank: A tangent to a circle intersects it in _____ point(s).

Solution. A tangent touches the circle at exactly one point, the point of contact. Every other point of the tangent lies outside the circle.

Common mistake. *A line that meets the circle in* two *points is not a tangent; it is a secant (see part (ii)).*

Answer: one.

Exercise 10.1, Question 2 (ii)

Question. Fill in the blank: A line intersecting a circle in two points is called a _____.

Solution. By definition, a line that cuts a circle at two distinct points is a secant of the circle.

Answer: secant.

Exercise 10.1, Question 2 (iii)

Question. Fill in the blank: A circle can have _____ parallel tangents at the most.

Solution. The tangents at the two ends of a diameter are parallel (both are perpendicular to that diameter). A tangent parallel to these must also be perpendicular to the same diameter at its point of contact, and the only such points on the circle are the two ends of the diameter. So there is no third parallel tangent.

Common mistake. *Having* many *tangents (part of Question 1) is different from having many* parallel *tangents. For one fixed direction there are exactly two tangents, one on each side of the circle.*

Answer: two.

Exercise 10.1, Question 2 (iv)

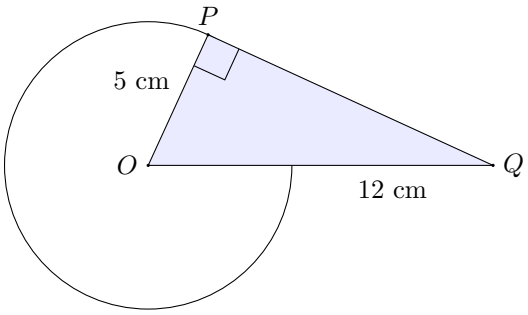
Question. Fill in the blank: The common point of a tangent to a circle and the circle is called _____.

Solution. This is the definition of the point of contact.

Answer: point of contact.

Exercise 10.1, Question 3

Question. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. Length PQ is:
(A) 12 cm (B) 13 cm (C) 8.5 cm (D) $\sqrt{119}$ cm.



Circle with centre O, tangent PQ at P, $OQ = 12$ cm. The shaded $\triangle OPQ$ has a right angle at P (tangent $\perp$ radius).

Solution. Join OP. The tangent is perpendicular to the radius through the point of contact, so $OP \perp PQ$, that is $\angle OPQ = 90°$. Hence $\triangle OPQ$ is a right-angled triangle with the right angle at P. Its hypotenuse is the side opposite the right angle, which is OQ; the two legs are $OP = 5$ cm and PQ.

By the Pythagoras theorem, (hypotenuse)^2 = (one leg)^2 + (other leg)^2, so $OQ^2 = OP^2 + PQ^2$:

$$\begin{aligned} PQ^2 &= OQ^2 - OP^2 \\ &= 12^2 - 5^2 \\ &= 144 - 25 = 119 \\ PQ &= \sqrt{119} \text{ cm.} \end{aligned}$$

Common mistake. *The right angle is at P, not at O, so the hypotenuse is $OQ = 12$. The squares of the two legs add up to 12^2, so we* subtract*: $PQ^2 = 12^2 - 5^2$. Adding, $12^2 + 5^2 = 169$, gives 13, which is option (B), a tempting wrong choice.*

Checking the other options. PQ is a leg of the right triangle, so $PQ < OQ = 12$ cm; this rules out (A) and (B). For (C), $8.5^2 = 72.25 \neq 119$.

Verification. Put $PQ^2 = 119$ back: $OP^2 + PQ^2 = 25 + 119 = 144 = 12^2 = OQ^2$.

Answer: Option (D): $PQ = \sqrt{119}$ cm.

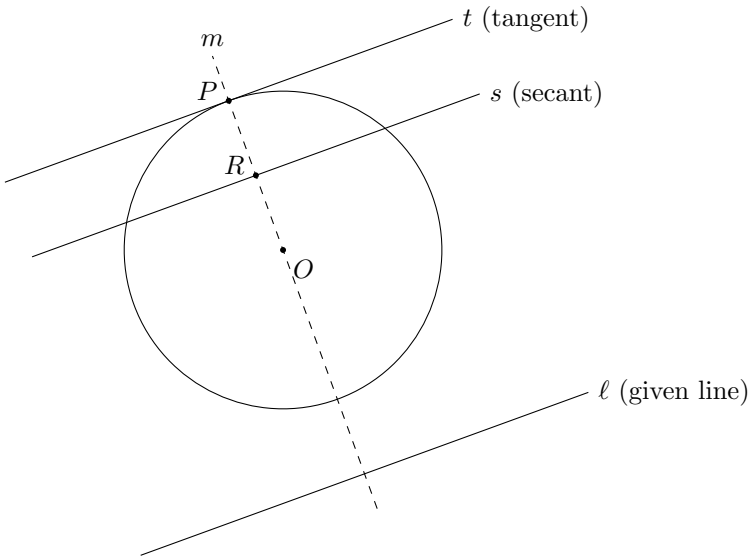
Exercise 10.1, Question 4

Question. Draw a circle and two lines parallel to a given line such that one is a tangent and the other, a secant to the circle.

Solution. This is a construction question. Unlike the other questions in this chapter, the figure here *is* the answer, so it shows the finished construction.

Steps of construction.

1. Draw the given line ℓ. Draw a circle with centre O and radius 3 cm.
2. Through O draw the line m perpendicular to ℓ (using a set square, or the compass construction for a perpendicular from a point to a line). Let m meet the circle at P.
3. At P draw the line t perpendicular to OP.
4. Mark a point R on OP with $OR = 1.5$ cm. Through R draw the line s perpendicular to m.



The tangent t and the secant s, both parallel to the given line ℓ.

Reasoning.

- ℓ, t and s are all perpendicular to the same line m, so they are parallel to one another.
- t is perpendicular to the radius OP at its end P on the circle. Every other point of t is the far end of a hypotenuse from O, so it is farther than OP from O and lies outside the circle. So t meets the circle only at P: it is a tangent.
- $OR = 1.5$ cm is less than the radius 3 cm, so R is inside the circle. Any line through an interior point cuts the circle at two points, so s is a secant.

Common mistake. *The tangent must be perpendicular to the* radius OP *at P, and it must pass through P on the circle. A line through P at any other angle to OP would cut the circle again and be a secant.*

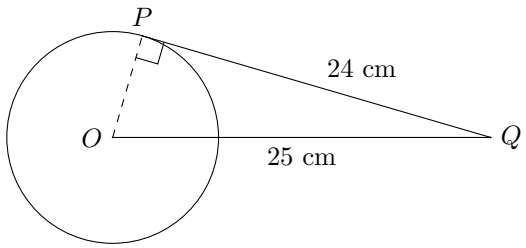
Answer: The line t (tangent) and the line s (secant) are both parallel to the given line ℓ.

Part C: Exercise 10.2

In Q.1 to 3, choose the correct option and give justification.

Exercise 10.2, Question 1

Question. From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is
(A) 7 cm (B) 12 cm (C) 15 cm (D) 24.5 cm



QP is the tangent from Q, P the point of contact, O the centre. Dashed line: construction.

Solution. Let the tangent from Q touch the circle at P, so $QP = 24$ cm and $OQ = 25$ cm. Join OP; it is the radius r.

The tangent is perpendicular to the radius through the point of contact, so $\angle OPQ = 90°$. In the right triangle OPQ, by the Pythagoras theorem $OQ^2 = OP^2 + PQ^2$:

$$\begin{aligned} OP^2 &= OQ^2 - PQ^2 \\ &= 25^2 - 24^2 \\ &= 625 - 576 = 49 \\ OP &= \sqrt{49} = 7 \text{ cm}. \end{aligned}$$

Common mistake. *The right angle is at P (where the tangent meets the radius), so the hypotenuse is $OQ = 25$, the distance from the centre. The radius is a* leg. *Using $OP^2 = 25^2 + 24^2$ would treat the radius as the hypotenuse, which is wrong.*

Verification. $OP^2 + PQ^2 = 7^2 + 24^2 = 49 + 576 = 625 = 25^2 = OQ^2$.

Answer: Option (A): the radius is 7 cm.

Exercise 10.2, Question 2

Question. In Fig. 10.11, if TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110°$, then $\angle PTQ$ is equal to
(A) 60° (B) 70° (C) 80° (D) 90°

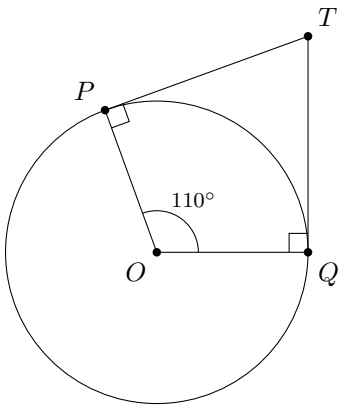


Fig. 10.11 (redrawn).

Solution. TP and TQ are tangents, so each is perpendicular to the radius through its point of contact (the tangent is perpendicular to the radius at the point of contact):

$$\angle OPT = 90°, \qquad \angle OQT = 90°.$$

The angles of the quadrilateral $OPTQ$ add up to 360°:

$$\begin{aligned} \angle OPT + \angle PTQ + \angle OQT + \angle POQ &= 360° \\ 90° + \angle PTQ + 90° + 110° &= 360° \\ \angle PTQ &= 360° - 290° = 70°. \end{aligned}$$

Here $90° + 90° + 110° = 290°$.

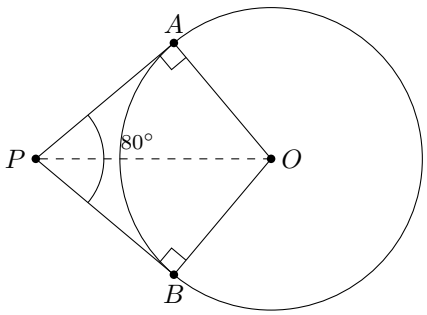
Common mistake. *The four angles of $OPTQ$ are the two right angles at P and Q, the given $\angle POQ = 110°$, and the unknown $\angle PTQ$. A common slip is to use* 180° *(triangle sum) instead of* 360° *(quadrilateral sum).*

Verification. $\angle PTQ + \angle POQ = 70° + 110° = 180°$, which agrees with the result of Question 10 (the angle between the tangents and the angle at the centre are supplementary).

Answer: Option (B): $\angle PTQ = 70°$.

Exercise 10.2, Question 3

Question. If tangents PA and PB from a point P to a circle with centre O are inclined to each other at angle of 80°, then $\angle POA$ is equal to
(A) 50° (B) 60° (C) 70° (D) 80°



Tangents PA and PB; dashed line: construction OP.

Solution. The tangent is perpendicular to the radius through the point of contact, so $\angle OAP = \angle OBP = 90°$.

In $\triangle OAP$ and $\triangle OBP$: $OA = OB$ (radii), OP is common, and $\angle OAP = \angle OBP = 90°$. So $\triangle OAP \cong \triangle OBP$ (RHS) and $\angle OPA = \angle OPB$ (CPCT). Therefore OP bisects $\angle APB$:

$$\angle OPA = \tfrac{1}{2} \times 80° = 40°.$$

In $\triangle OAP$, by the angle-sum property,

$$\angle POA = 180° - \angle OAP - \angle OPA = 180° - 90° - 40° = 50°.$$

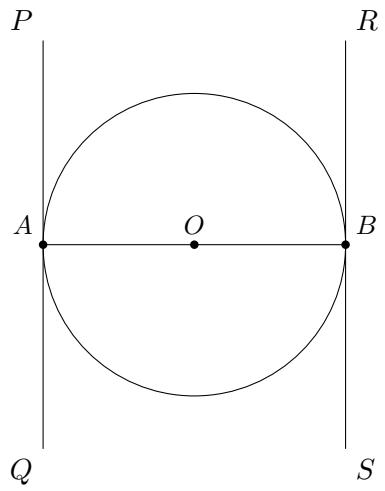
Common mistake. *The* $80°$ *is the angle* $\angle APB$ between the tangents*, not* $\angle POA$*. The line* OP *cuts it into two halves of* $40°$*. Answering* $80°$ *(option D) confuses that angle with* $\angle POA$*; stopping at* $\angle OPA = 40°$ *forgets the final angle-sum step in* $\triangle OAP$*.*

Verification. By Question 10, $\angle AOB = 180° - 80° = 100°$, and OP bisects it, so $\angle POA = \tfrac{1}{2} \times 100° = 50°$.

Answer: Option (A): $\angle POA = 50°$.

Exercise 10.2, Question 4

Question. Prove that the tangents drawn at the ends of a diameter of a circle are parallel.



AB is a diameter; PQ and RS are the tangents at A and B.

Solution.

Given: A circle with centre O and diameter AB. PQ is the tangent at A and RS is the tangent at B, with P and R on the same side of AB.

To prove: $PQ \parallel RS$.

Proof:

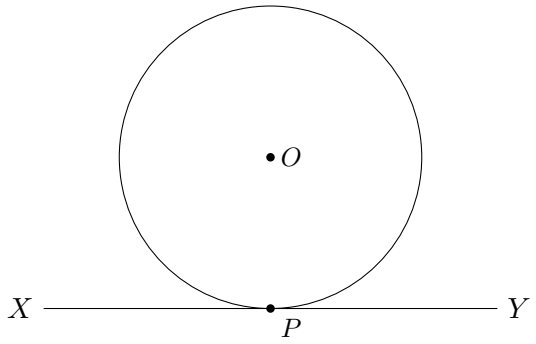
Step	Statement	Reason
1	$\angle PAB = 90°$.	The tangent at A is perpendicular to the radius OA, and OA lies along the diameter AB.

Step	Statement	Reason (continued)
2	$\angle ABS = 90°$.	The tangent at B is perpendicular to the radius OB, and OB lies along AB.
3	$\angle PAB = \angle ABS$.	Each is 90° (steps 1 and 2).
4	These are alternate interior angles formed by the lines PQ and RS with the transversal AB.	P and S lie on opposite sides of AB.
5	$PQ \parallel RS$.	If a transversal makes equal alternate interior angles with two lines, the lines are parallel.

Answer: The tangents at the ends of a diameter are parallel. (Proved)

Exercise 10.2, Question 5

Question. Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre.



XY is the tangent at P; O is the centre.

Solution.

Given: A circle with centre O and a tangent XY touching it at P. Let ℓ be the line through P perpendicular to XY.

To prove: ℓ passes through O.

Construction: Join OP.

Proof:

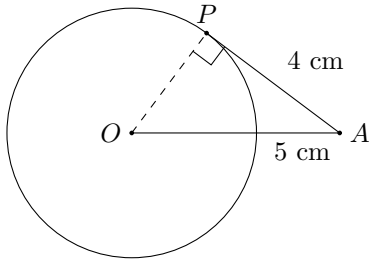
Step	Statement	Reason
1	$OP \perp XY$.	The tangent is perpendicular to the radius through the point of contact.
2	$\ell \perp XY$ at P.	Given.
3	OP and ℓ are both perpendicular to XY at the same point P.	Steps 1 and 2.
4	OP and ℓ are the same line.	Through a given point of a line, only one perpendicular to that line can be drawn.

Step	Statement	Reason (continued)
5	ℓ passes through O.	OP passes through O (step 4).

Answer: The perpendicular to a tangent at the point of contact passes through the centre of the circle. (Proved)

Exercise 10.2, Question 6

Question. The length of a tangent from a point A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.



AP is the tangent from A with point of contact P. Dashed line: construction.

Solution. Let the tangent from A touch the circle at P, so $AP = 4$ cm and $OA = 5$ cm. Join OP; it is the radius r.

The tangent is perpendicular to the radius through the point of contact, so $\angle OPA = 90°$. By the Pythagoras theorem in $\triangle OPA$, $OA^2 = OP^2 + AP^2$:

$$\begin{aligned} OP^2 &= OA^2 - AP^2 \\ &= 5^2 - 4^2 \\ &= 25 - 16 = 9 \\ OP &= \sqrt{9} = 3 \text{ cm}. \end{aligned}$$

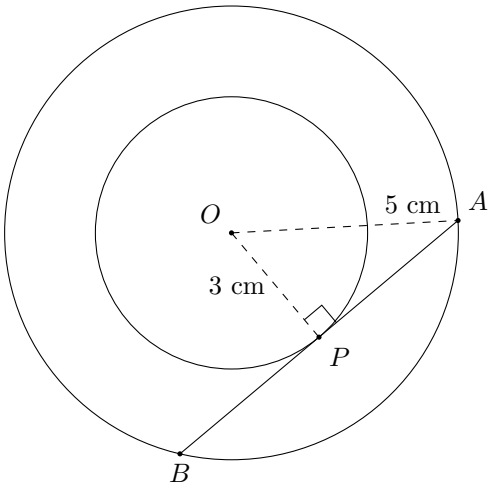
Common mistake. *The distance $OA = 5$ cm from the centre is the hypotenuse (opposite the right angle at P), and the tangent length $AP = 4$ cm and the radius are the legs. So subtract: $5^2 - 4^2$, not $5^2 + 4^2$.*

Verification. $OP^2 + AP^2 = 3^2 + 4^2 = 9 + 16 = 25 = 5^2 = OA^2$.

Answer: The radius of the circle is 3 cm.

Exercise 10.2, Question 7

Question. Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.



Chord AB of the larger circle touches the smaller circle at P. Dashed lines: construction.

Solution. Let AB be the chord of the larger circle (radius 5 cm) that touches the smaller circle (radius 3 cm) at P. Join OP and OA, so $OP = 3$ cm and $OA = 5$ cm.

AB is a tangent to the smaller circle at P, so $OP \perp AB$ (the tangent is perpendicular to the radius through the point of contact). In the right triangle OPA, by the Pythagoras theorem $OA^2 = OP^2 + AP^2$:

$$\begin{aligned} AP^2 &= OA^2 - OP^2 \\ &= 5^2 - 3^2 \\ &= 25 - 9 = 16 \\ AP &= \sqrt{16} = 4 \text{ cm}. \end{aligned}$$

OP is the perpendicular from the centre O to the chord AB of the larger circle, so it bisects the chord (the perpendicular from the centre of a circle to a chord bisects the chord; see Example 1). Hence $AB = 2 \times AP = 2 \times 4 = 8$ cm.

Common mistake. *$AP = 4$ cm is only* half *the chord. The question asks for the whole chord AB, so the last step 2×4 must not be left out.*

Verification. Half-chord $AP = 4$, so $AP^2 + OP^2 = 4^2 + 3^2 = 16 + 9 = 25 = 5^2 = OA^2$.

Answer: The length of the chord is 8 cm.

Exercise 10.2, Question 8

Question. A quadrilateral $ABCD$ is drawn to circumscribe a circle (see Fig. 10.12). Prove that $AB + CD = AD + BC$.

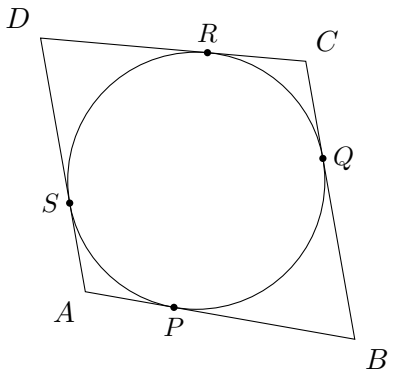


Fig. 10.12 (redrawn). P, Q, R, S are the points of contact on AB, BC, CD, DA.

Solution.

Given: A quadrilateral $ABCD$ whose sides AB, BC, CD, DA touch a circle at P, Q, R, S respectively.

To prove: $AB + CD = AD + BC$.

Proof:

Step	Statement	Reason
1	$AP = AS$	Tangents from the external point A are equal.
2	$BP = BQ$	Tangents from the external point B are equal.
3	$CR = CQ$	Tangents from the external point C are equal.
4	$DR = DS$	Tangents from the external point D are equal.
5	$AP+BP+CR+DR = AS+BQ+CQ+DS$	Adding steps 1 to 4.
6	$(AP + BP) + (CR + DR) =$ $(AS + DS) + (BQ + CQ)$	Regrouping step 5.
7	$AB + CD = AD + BC$	P lies on AB, R on CD, S on DA, Q on BC, so $AP + BP = AB$, $CR + DR = CD$, $AS + DS = AD$, $BQ + CQ = BC$.

Answer: $AB + CD = AD + BC$. (Proved)

Exercise 10.2, Question 9

Question. In Fig. 10.13, XY and $X'Y'$ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and $X'Y'$ at B. Prove that $\angle AOB = 90°$.

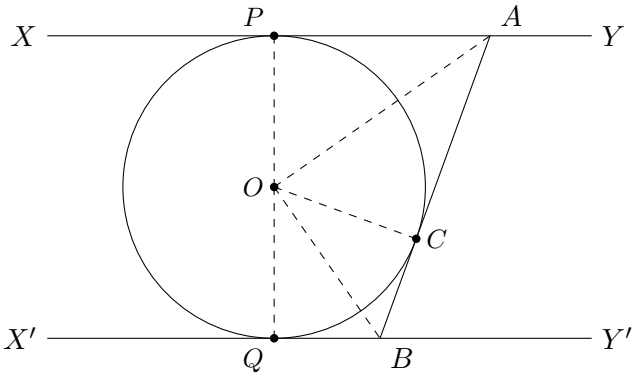


Fig. 10.13 (redrawn). P and Q are the points of contact of XY and $X'Y'$. Dashed lines: construction.

Solution.

Given: $XY \parallel X'Y'$ are tangents to a circle with centre O, touching it at P and Q. The tangent AB touches the circle at C and meets XY at A and $X'Y'$ at B.

To prove: $\angle AOB = 90°$.

Construction: Join OP, OQ, OC, OA and OB.

Proof:

Step	Statement	Reason
1	$\angle OPA = \angle OCA = 90°$.	The tangent is perpendicular to the radius through the point of contact.
2	In $\triangle OPA$ and $\triangle OCA$: $OP = OC$, OA is common and $\angle OPA = \angle OCA = 90°$. So $\triangle OPA \cong \triangle OCA$.	OP, OC are radii; RHS congruence.
3	$\angle OAP = \angle OAC$, so $\angle PAB = 2\angle OAB$.	CPCT; C lies on AB, so $\angle OAC = \angle OAB$ and $\angle PAC = \angle PAB$.
4	Similarly $\triangle OQB \cong \triangle OCB$, so $\angle QBA = 2\angle OBA$.	Same reasoning at B (RHS, then CPCT).
5	$\angle PAB + \angle QBA = 180°$.	Interior angles on the same side of the transversal AB for the parallel lines XY and $X'Y'$.
6	$2\angle OAB + 2\angle OBA = 180°$, so $\angle OAB + \angle OBA = 90°$.	Steps 3, 4 and 5.
7	$\angle AOB = 180° - (\angle OAB + \angle OBA) = 90°$.	Angle-sum property of $\triangle AOB$.

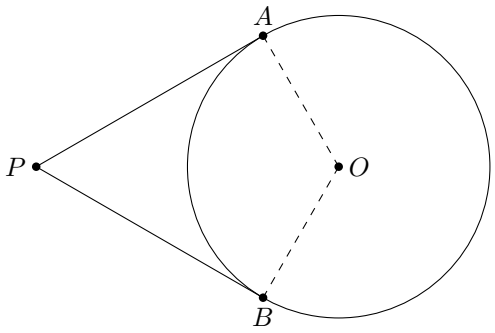
Common mistake. *Step 3 says $\angle PAB = 2\angle OAB$, not $\angle PAB = \angle OAB$: AO bisects the angle $\angle PAB$ because the two triangles at A are congruent. Forgetting the factor 2 makes step 6 give $\angle OAB + \angle OBA = 180°$, which is wrong.*

Answer: $\angle AOB = 90°$. (Proved)

Alternative proof. Join OP, OQ, OC. By RHS, $\triangle OPA \cong \triangle OCA$ and $\triangle OQB \cong \triangle OCB$, so $\angle AOP = \angle AOC$ and $\angle BOQ = \angle BOC$. Since $XY \parallel X'Y'$, both OP and OQ are perpendicular to $X'Y'$ through O, so POQ is a straight line and $\angle POQ = 180°$. Then $\angle AOP + \angle AOC + \angle BOC + \angle BOQ = 180°$, i.e. $2(\angle AOC + \angle BOC) = 180°$, so $\angle AOB = \angle AOC + \angle BOC = 90°$.

Exercise 10.2, Question 10

Question. Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre.



PA and PB are tangents from P; A and B are the points of contact. Dashed lines: construction.

Solution.

Given: A circle with centre O; an external point P; tangents PA and PB touching the circle at A and B.

To prove: $\angle APB + \angle AOB = 180°$.

Construction: Join OA and OB.

Proof:

Step	Statement	Reason
1	$\angle OAP = 90°$ and $\angle OBP = 90°$.	The tangent is perpendicular to the radius through the point of contact.
2	$\angle OAP + \angle APB + \angle OBP + \angle AOB = 360°$.	Angle-sum property of the quadrilateral $OAPB$.
3	$90° + \angle APB + 90° + \angle AOB = 360°$.	Substituting step 1 in step 2.
4	$\angle APB + \angle AOB = 180°$.	From step 3.

Answer: The angle between the two tangents and the angle subtended at the centre by the segment joining the points of contact add up to 180°, i.e. they are supplementary. (Proved)

Exercise 10.2, Question 11

Question. Prove that the parallelogram circumscribing a circle is a rhombus.

Note on the figure. No separate figure is drawn: an accurate figure of a parallelogram circumscribing a circle is already a rhombus, so it would show the result. Use the labels of the figure of Question 8 (points of contact P, Q, R, S on AB, BC, CD, DA).

Solution.

Given: A parallelogram $ABCD$ whose sides AB, BC, CD, DA touch a circle at P, Q, R, S.

To prove: $ABCD$ is a rhombus, i.e. $AB = BC = CD = DA$.

Proof:

Step	Statement	Reason
1	$AB + CD = AD + BC$.	For a quadrilateral circumscribing a circle, sums of opposite sides are equal (Question 8, which uses "tangents from an external point are equal").
2	$AB = CD$ and $AD = BC$.	Opposite sides of a parallelogram are equal.
3	$2AB = 2AD$, so $AB = AD$.	Substituting step 2 in step 1.
4	$AB = BC = CD = DA$.	$AB = CD$, $AD = BC$ (step 2) and $AB = AD$ (step 3).
5	$ABCD$ is a rhombus.	A parallelogram with all four sides equal is a rhombus.

Answer: A parallelogram that circumscribes a circle is a rhombus. (Proved)

Exercise 10.2, Question 12

Question. A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 8 cm and 6 cm respectively (see Fig. 10.14). Find the sides AB and AC.

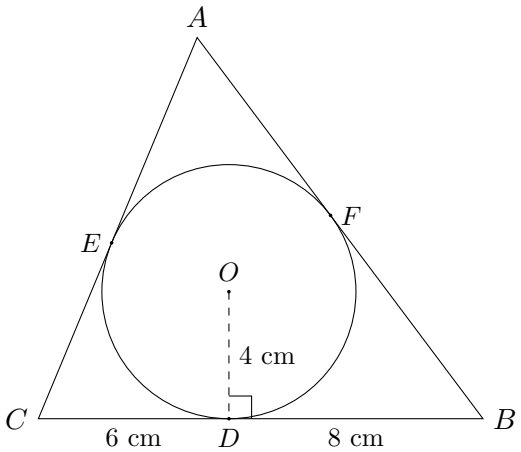


Fig. 10.14 (redrawn). The circle touches BC, CA, AB at D, E, F.

Solution. Let the circle touch BC, CA, AB at D, E, F respectively, and let $AE = AF = x$ cm.

Step 1: express the sides using equal tangents (tangents from an external point are equal):

$$BF = BD = 8 \text{ cm}, \qquad CE = CD = 6 \text{ cm}, \qquad AF = AE = x \text{ cm},$$
$$AB = AF + FB = (x+8) \text{ cm}, \quad AC = AE + EC = (x+6) \text{ cm}, \quad BC = BD + DC = 14 \text{ cm}.$$

Step 2: area by Heron's formula. For sides a, b, c and $s = \dfrac{a+b+c}{2}$,

$$\text{area} = \sqrt{s(s-a)(s-b)(s-c)}.$$

Take $a = BC = 14$, $b = CA = x+6$, $c = AB = x+8$:

$$s = \frac{14 + (x+6) + (x+8)}{2} = x + 14, \quad s - a = x, \quad s - b = 8, \quad s - c = 6.$$

$$\text{ar}(\triangle ABC) = \sqrt{(x+14) \cdot x \cdot 8 \cdot 6} = \sqrt{48x(x+14)}.$$

Step 3: area using the radius. Join O to A, B, C. The triangle ABC splits into $\triangle OBC$, $\triangle OCA$, $\triangle OAB$. Each has height equal to the radius $r = 4$ cm, because the radius to a point of contact is perpendicular to the tangent. Area of a triangle with inradius r is $\frac{1}{2} \times r \times$ perimeter:

$$\text{ar}(\triangle ABC) = \tfrac{1}{2} \times 4 \times \big(14 + (x+6) + (x+8)\big) = \tfrac{1}{2} \times 4 \times (2x + 28) = 4(x+14).$$

Step 4: equate and solve. Both expressions are the area of $\triangle ABC$:

$$\begin{aligned}
\sqrt{48x(x+14)} &= 4(x+14) \\
48x(x+14) &= \big(4(x+14)\big)^2 = 16(x+14)^2 && \text{(squaring both sides)} \\
\frac{48x(x+14)}{16(x+14)} &= \frac{16(x+14)^2}{16(x+14)} && \text{(dividing both sides by } 16(x+14) \neq 0\text{)} \\
3x &= x + 14 && \text{(}48 \div 16 = 3 \text{ and } (x+14)^2 \div (x+14) = x+14\text{)} \\
3x - x &= 14 \\
2x &= 14 \\
x &= 7.
\end{aligned}$$

Therefore, putting $x = 7$: $AB = x + 8 = 7 + 8 = 15$ cm and $AC = x + 6 = 7 + 6 = 13$ cm.

Common mistake. *In Step 4 the factor $(x+14)$ appears on both sides, so dividing by it is allowed only because $x + 14 \neq 0$ (lengths are positive). Also, $AB = x + 8$ and $AC = x + 6$: the 8 goes with B and the 6 with C, because $BD = 8$ and $CD = 6$.*

Verification. The sides are $AB = 15$, $AC = 13$, $BC = 14$, so $s = \dfrac{15 + 13 + 14}{2} = 21$. By Heron's formula:

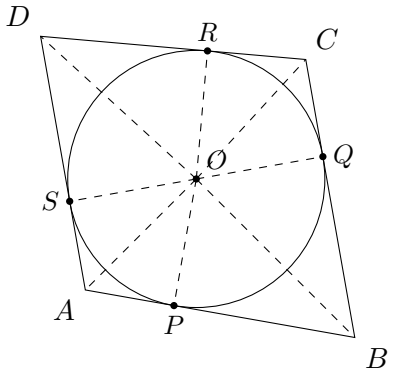
$$\text{area} = \sqrt{21 \cdot (21-14)(21-13)(21-15)} = \sqrt{21 \cdot 7 \cdot 8 \cdot 6} = \sqrt{7056} = 84.$$

By the inradius formula: $\frac{1}{2} \times 4 \times (15 + 13 + 14) = \frac{1}{2} \times 4 \times 42 = 84$. The two areas agree.

Answer: $AB = 15$ cm and $AC = 13$ cm.

Exercise 10.2, Question 13

Question. Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.



$ABCD$ circumscribes the circle with centre O; P, Q, R, S are the points of contact. Dashed lines: construction.

Solution.

Given: A quadrilateral $ABCD$ circumscribing a circle with centre O; the sides AB, BC, CD, DA touch the circle at P, Q, R, S.

To prove: $\angle AOB + \angle COD = 180^\circ$ and $\angle AOD + \angle BOC = 180^\circ$.

Construction: Join OA, OB, OC, OD, OP, OQ, OR, OS.

Proof:

Step	Statement	Reason
1	$\angle OPA = \angle OSA = 90^\circ$.	The tangent is perpendicular to the radius through the point of contact.
2	In $\triangle OPA$ and $\triangle OSA$: $OP = OS$, OA is common, $\angle OPA = \angle OSA = 90^\circ$. So $\triangle OPA \cong \triangle OSA$.	OP, OS are radii; RHS congruence.
3	$\angle AOP = \angle AOS$. Call each a.	CPCT.
4	In the same way, $\angle BOP = \angle BOQ = b$, $\angle COQ = \angle COR = c$, $\angle DOR = \angle DOS = d$.	RHS and CPCT at B, C, D.
5	$2a + 2b + 2c + 2d = 360^\circ$, so $a + b + c + d = 180^\circ$.	The angles around the point O add up to 360°.
6	$\angle AOB = a + b$, $\angle COD = c + d$, $\angle AOD = a + d$, $\angle BOC = b + c$.	OP lies inside $\angle AOB$, OQ inside $\angle BOC$, OR inside $\angle COD$, OS inside $\angle DOA$.
7	$\angle AOB + \angle COD = (a + b) + (c + d) = 180^\circ$.	Steps 5 and 6.
8	$\angle AOD + \angle BOC = (a + d) + (b + c) = 180^\circ$.	Steps 5 and 6.

Answer: Opposite sides AB, CD subtend supplementary angles at O, and so do AD, BC. (Proved)